

Exp :- 5

Aim :- Measurement of primary and Secondary Voltage & Current and power in Single phase Transformer.

Required Apparatus :-

Sno	Material Required	Specification Rating	quantity
(1)	Transformer Panel	1- Phase	1
(2)	Variable Resistive Load	0 - 1500W	1
(3)	Connecting Leads	1mm Square	As per Req.

Theory :- The transformer is the static device which work on the principal of electromagnetic Induction. It is used for Transferring the electrical Power from one circuit to another circuit without any variation in their frequency.

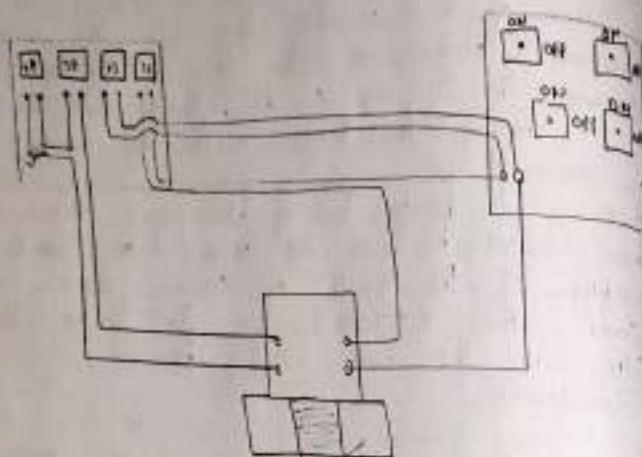
Working of Transformer :- Transformer one winding (also known as a coil) is supplied by an alternating electrical source. The alternating current through the winding produce a Continuously changing and alternating flux that surrounds the winding.

If another winding is brought close to winding, some portion of this alternating flux will link with secondary winding. As the flux is continually changing in its amp & dirⁿ, there must be changing flux linkage in secondary winding or coil.

According to Faraday's law of electromagnetic induction there will be an EMF induced in second winding. If the circuit of secondary winding is closed, then a current will flow through it. This is the basic working principle of Transformer.

Parts of Transformer :- There are 3 main parts

- Primary winding of Transformer : It produces magnetic flux when it is connected to an AC electric source.
- Magnetic Core Transformer : The magnetic flux produced by primary winding, that will pass through this low reluctance path linked with secondary winding & create a close magnetic circuit.
- Secondary winding of Transformer : The flux, produced by primary winding, passes through the core, will link with the secondary.



Single phase Transformer

winding. This winding also winds on the same core and give the desired output to load.

Losses in Transformer: A types of losses are.

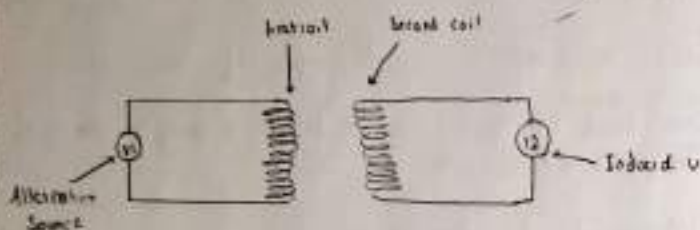
→ Core losses or Iron losses: losses occurring in core of Transformer. These are.

→ Hysteresis loss in Transformer:

Hysteresis loss occurs when there is reversal in the process of magnetization in the Transformer core. This loss depends upon the volume and grade of iron, value of flux density and freq. of mag. reversals.

→ Eddy current loss in Transformer

In Transformer, AC is supplied to the primary winding it produce set up alternating magnetizing flux. When this flux links with secondary winding.



~~Flux~~

Copper loss : Copper loss occurs due to resistance of windings of the transformer for primary winding, it is $I_1^2 R_1$ and for secondary winding, it is $I_2^2 R_2$ where I_1 & I_2 are primary & secondary current respectively, R_1 & R_2 are primary winding resistance and secondary winding resist. Repet

Ideal Transformer : An Ideal Transformer is a transformer which has

- No copper loss
- No iron loss
- No leakage flux
- 100% efficiency

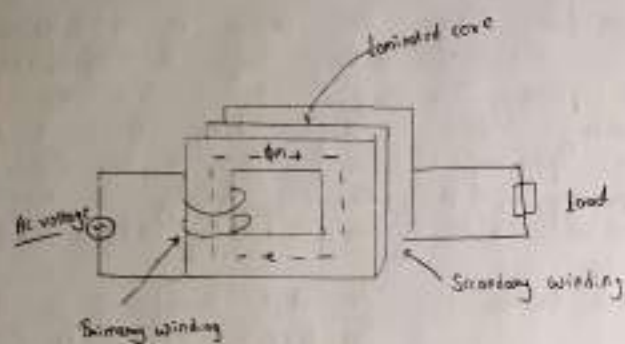
Relation of Voltage, current and no. of Turns of winding

$$\frac{V_1}{V_2} = \frac{N_1}{N_2} = \frac{I_2}{I_1}$$

V_1 & V_2 = voltage across Pri, sec winding.

I_1 & I_2 = current winding in Pri, Sec

N_1 & N_2 = No of Turns in Pri & Sec winding



~~Electrical Transformer~~

Types of Transformer

Set up and Set down Transformer

Step up - $V_1 < V_2$ OR $N_1 < N_2$ OR $I_1 < I_2$
 Step down - $V_1 > V_2$ OR $N_1 > N_2$ OR $I_1 > I_2$

Observation Table

Load	V_P	I_P	V_S	I_S
No load	100	0.01	41	0.00
Load 1	91	0.93	96	1.87
Load 2	93	0.74	96	1.56

Result

$\therefore$ We have measured the primary & secondary voltage, currents, single phase transformer with different loading conditions.

Viva questions $\rightarrow$

Viva question

(1) What is Transformer?

It is a Static device which is used to change the level of voltage or current without changing the frequency & power.

(2) What do you mean by Turns ratio of Transformer.

= It is Ratio of primary Turns & Secondary Turns

(3) What is Transformation Ratio of Transformer?

= Ratio of Secondary Side Turns to primary Side T.

(4) Types of Transformer?

= Step up T and Step down Transformer.

(5) Why ~~core~~ of Transformer is laminated?

= To reduce eddy current loss

(6) Ideal Transformer?

= Transformation which has no losses and 100 % efficiency.

(7) What is Copper losses in Transformer?

= Heat loss in winding of Transformer due to flow of current in these.

(8) What is Iron loss in Transformer?

= Heat loss in Core of Transformer.

(9) What are different polarities of Transformer?

= Positive and negative.

(10) What is working principle of autoTransformer?

= Electromagnetic Induction.

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10/4/24